INCH-POUND
MIL-DTL-24643/56E
22 August 2002
SUPERSEDING
MIL-C-24643/56D
22 November 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS2SWL-7

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (watertight)

First -	Copper conductor, tin coated, AWG size 16, class B stranding, ASTM B 8.
Second -	Separator (opaque white) may be used at manufacturer's option where required to provide free-stripping insulation.
Third -	Cross-linked polyethylene insulation. Nominal wall thickness 0.019 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. Colored insulation, one black and one white conductor for each pair.
Fourth -	Two conductors (one black and one white) cabled together with a nominal lay of 3 inches to form a pair.
Fifth -	Binder tape over each pair, applied at manufacturer's option, applied helically with overlap.
Sixth -	Braided shield of AWG No. 34 or No. 36 tin coated copper. Braid angle of 30 to 35 degrees; minimum coverage of 85 percent.

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Construction (watertight) (continued)

Seventh - Shield insulation of two polyester tapes, per NEMA FI-4 outer tape sealed. An alternate shield insulation may consist of one polyester tape per NEMA FI-4 plus an extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2 on the inner tape.

Eighth - Seven shielded pairs cabled together with a lay not greater than 18 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with pair No. 1, from the center outward. Fillers shall be used as necessary to make a firm, well-rounded assembly.

Ninth - Binder tape, applied helically with overlap.

Tenth - Cross-linked polyolefin jacket. Minimum average jacket thickness 0.075 inch. Cable overall diameter shall be 0.860 minimum, 0.910 maximum inch.

NOTES:

1. LS2SWL-7 military part number – M24643/56-01UN.

EXAMINATION AND TESTS:

Basic electrical:	Requirements
Conductor Resistance - ohms/1000 feet at 25°C, maximum	4.66
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Conductor to shield	1000
Shield to shield (pair shields only)	500
Insulation resistance - Megohms -1000 feet minimum	
Conductor to shield	500
Component shield to shield	100
Component and shield continuity	No failure
Group A:	
Visual and dimensional	No failure
Watertightness - see MIL-DTL-24643 for water leakage	No failure
Capacitance	
Mutual capacitance - 1 kilohertz (kHz), pF/ft, maximum	35
Capacitance unbalance - percent maximum	10
Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Drip - 95 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35

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Physicals (unaged)
- Insulation
  - Tensile strength - lb/in^2, minimum 1800
  - Elongation - percent, minimum 250
- Jacket (cable)
  - Tensile strength - lb/in^2, minimum 1300
  - Elongation - percent, minimum 160

Group C:

Physicals - (aged)
- Insulation
  - Air oven
    - Tensile strength - percent of unaged, minimum 80
    - Elongation - percent of unaged minimum 80
- Jacket (cable)
  - Air oven
    - Tensile strength - percent of unaged, minimum 60
    - Elongation - percent of unaged minimum 60
  - Hot oil immersion
    - Tensile strength - percent of unaged, minimum 50
    - Elongation - percent of unaged, minimum 50
- Shrinkage ..... No failure
- Heat distortion - percent of unaged, maximum 30
- Permanence of printing (jacket) - cycles, minimum 125
- Cable sealant removability ..... No failure
- Shield conformance to material, construction and coverage ..... No failure

Group D:
- Flame propagation (cable) ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

- Aging and compatibility (cable) (125 ± 5°C) ..... No failure
- Abrasion resistance (jacket) - scrapes, minimum 75
- Acid gas equivalent - percent, maximum
  - Jacket 2
  - Fillers 2
  - Insulation 18

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Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS: 1500 feet nominal

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-054)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

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